

Objective	Platform	Program/Function	Infiles	Command	Outfiles	Other
Preprocessing of sequence reads						
Check raw read stats	Ubuntu/Unix	grep	*fastq.gz	zgrep -c @M03353 *fastq.gz > Latnja-browning-Metagenomics-raw-sequence-count.txt	Latnja-browning-Metagenomics-raw-sequence-count.txt	
Quality check on raw sequences	Ubuntu/Unix	fastqc	*fastq.gz	fastqc -t 2 *fastq.gz -o ~/NGS/Latnja/Browning/Metagenomics/fastqc_out	folder with multiple output files to view in web browser	
Rename files	Ubuntu/Unix	mv	*fastq.gz	for i in \$(ls *.gz); do mv "\$i" "\${i/_S*[0-9]_L001/}"; done	*fastq.gz	
Trim reads	Ubuntu/Unix	Trimmomatic	*.fastq.gz	see bash script helper_scripts/trimmomatic_automate.sh	*_trimmed.fastq	
Check trimmed read stats	Ubuntu/Unix	grep	*fastq.gz	zgrep -c @M03353 *fastp* > Latnja-browning-Metagenomics-trimmed-sequence-count.txt	*_unpaired.fastq	
count reads total	Ubuntu/Unix	zcat, grep, wc	*fastq.gz	zcat *.gz grep '@' wc -l	output in console	Sequencing generated a total of 24764579 reads when all files are considered
Quality check on trimmed sequences	Ubuntu/Unix	fastqc	*fastq.gz	fastqc for base in BC1 BC2 BC3 BC4 BC5 BC6 BC7 BE1 BE5 BE7 HC2 HC3 HC4 HC6 HE1 HE5 HE6 HE7 do cat \$(base)_R1_trimmed.fastq \$(base)_R1_trimmed.fastq > \$(base)_combined_R1.fastq cat \$(base)_R2_trimmed.fastq \$(base)_R2_trimmed.fastq > \$(base)_combined_R2.fastq done	*_combined_R1.fastq *_combined_R2.fastq	
Pool technical replicates	Ubuntu/Unix	cat	*_trimmed.fastq	for f in *.fastq; do echo -n "\$f: " echo \$((\$(wc -l < "\$f") / 4)) done > brownning_combined_sequence_counts.txt	browning_combined_sequence_counts.txt	
count reads total	Ubuntu/Unix	echo, wc	*_combined_R1.fastq *_combined_R2.fastq	see script contamination_check.sh	*combined_R1_nohuman.fastq *combined_R2_nohuman.fastq	
Screen for contaminant sequences	Dardel	samtools, bowtie2	*_combined_R1.fastq *_combined_R2.fastq	for f in *.fastq; do echo -n "\$f: " echo \$((\$(wc -l < "\$f") / 4)) done > brownning_combined_nohuman_sequence_counts.txt		
count reads total	Ubuntu/Unix	echo, wc	*_R1_nohuman.fastq *_R2_nohuman.fastq		browning_combined_nohuman_sequence_counts.txt	
Taxonomy (Not used in the manuscript)						
Kraken	Dardel/Unix		*_R1_nohuman.fastq *_R2_nohuman.fastq	see script kraken_plusfp.sh OR kraken_gtdb.sh	*_combined_kraken_metagen_gtdb_report.tsv *_combined_kraken_metagen_gtdb_output.txt	Not used in the final manuscript. We rely instead on the 16S and ITS2 amplicon sequencing data for taxonomy details.
Bracken	Dardel/Unix		*_combined_R1.fastq *_combined_R2.fastq *report.tsv	see script bracken.sh OR bracken_gtdb.sh	*_combined_kraken_metagen_bracken_browning_F.tsv *_combined_kraken_metagen_bracken_browning_G.tsv *_combined_kraken_metagen_bracken_browning_S.tsv	Not used in the final manuscript. We rely instead on the 16S and ITS2 amplicon sequencing data for taxonomy details.
Analysis	RStudio		*_combined_kraken_metagen_bracken_browning_F.tsv *_combined_kraken_metagen_bracken_browning_G.tsv *_combined_kraken_metagen_bracken_browning_S.tsv	see script Browning_taxonomic_profiles_read_based.R	browning_combined_bracken_plusfp_annotated.csv browning_combined_bracken_gtdb.csv	These csv files are generated once and then used to re-run the downstream analyses in this script. This script and its output are not used in the final manuscript. We rely instead on the 16S and ITS2 amplicon sequencing data for taxonomy details.
Functional Annotation						
eggNOG	Dardel/Unix	eggNOG	*_combined_R1.fastq *_combined_R2.fastq	see script eggnog.sh	*_eggNOG_emapper *_eggNOG_emapper.genepred.fasta *_eggNOG_emapper.genepred.gff *_eggNOG_emapper.hits *_eggNOG_emapper.seed_orthologs	
Make new files for each type of functional information - CAZy, KEGG_ko, KEGG_Pathway, COG_category, PFAMs, EC	Ubuntu/Unix			see bash script extract_eggNOG_columns_function.sh	ko_list_clean.tsv metagenomic_browning_CAZy_counts.tsv metagenomic_browning_COG_category_counts.tsv metagenomic_browning_EC_counts.tsv metagenomic_browning_KEGG_ko_counts.tsv metagenomic_browning_KEGG_pathway_counts.tsv metagenomic_browning_PFAMs_counts.tsv	
Clean up KO_Orthology.txt from website	Ubuntu/Unix			awk -F"\t" 'BEGIN { OFS=" "; print "Main_code","Main_cat","Broad_cat","Broad_cat","Path_code","Path_name","Path_ko","KO_ID","Gene_desc" } { match(\$3, /^[0-9]+-+ (.+) \ PATH:(ko[0-9]+)\ \$/ , a) path_code=a[1] path_name=a[2] path_ko=a[3] ko_id=\$4 desc=\$5 for (i=6;i<NF;i++) desc=desc " \$i ec="" if (match(desc, /\[EC:[*]+\]/)) { ec=substr(desc,RSTART+4,RLENGTH-5) gsub(/\[EC:[*]+\]/,"", desc) } print \$1,\$2,path_code,path_name,path_ko,ko_id,desc,ec }' KO_Orthology.txt > KO_Orthology_clean.tsv	KO_Orthology_clean.tsv	
Analysis	RStudio			See R scripts: browning_functional_profiles_kegg.R browning_functional_profiles_cazy.R browning_functional_profiles_path_abund_and_cov.R		